

Q1 - 24 January - Shift 1

Let a tangent to the Curve $9x^2 + 16y^2 = 144$ intersect the coordinate axes at the points A and B. Then, the minimum length of the line segment AB is

Space for your notes:

Q2 - 24 January - Shift 1

Let C be the largest circle centred at (2, 0) and inscribed in the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If (1, α) lies on C, then $10\alpha^2$ is equal to _____

Space for your notes:

Q3 - 29 January - Shift 2

If the tangent at a point P on the parabola $y^2 = 3x$ is parallel to the line $x + 2y = 1$ and the tangents at the points Q and R on the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ are perpendicular to the line $x - y = 2$, then the area of the triangle PQR is:

Space for your notes:

- (1) $\frac{9}{\sqrt{5}}$ (2) $5\sqrt{3}$
(3) $\frac{3}{2}\sqrt{5}$ (4) $3\sqrt{5}$

Q4 - 31 January - Shift 1

If the maximum distance of normal to the ellipse $\frac{x^2}{4} + \frac{y^2}{b^2} = 1$, $b < 2$, from the origin is 1, then the eccentricity of the ellipse is:

(1) $\frac{1}{\sqrt{2}}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{2}$

(4) $\frac{\sqrt{3}}{4}$

Space for your notes:

Q5 - 01 February - Shift 2

The line $x = 8$ is the directrix of the ellipse

$E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with the corresponding focus

$(2, 0)$. If the tangent to E at the point P in the first quadrant passes through the point $(0, 4\sqrt{3})$ and intersects the x -axis at Q , then $(3PQ)^2$ is equal to _____.

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Answer Key

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(As per Official NTA Key released on 2 Feb)

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Q1 (7)

Q2 (118)

Q3 (4)

Q4 (2)

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Q5 (39)

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Q1 (7)

Equation of tangent at point $P(4 \cos \theta, 3 \sin \theta)$ is

$$\frac{x \cos \theta}{4} + \frac{y \sin \theta}{3} = 1 \text{ So A is } (4 \sec \theta, 0) \text{ and}$$

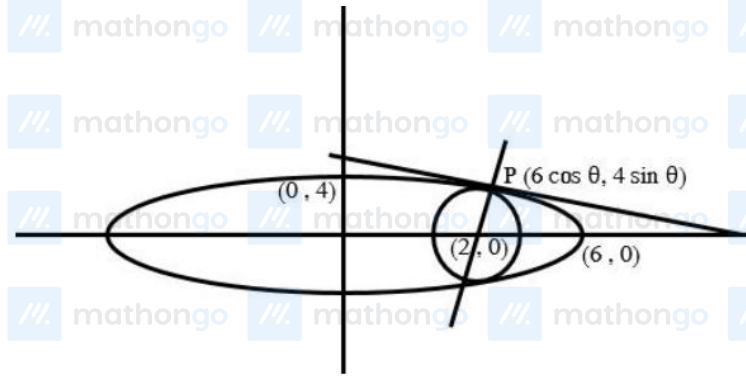
point B is $(0, 3 \operatorname{cosec} \theta)$

$$\text{Length AB} = \sqrt{16 \sec^2 \theta + 9 \operatorname{cosec}^2 \theta}$$

$$= \sqrt{25 + 16 \tan^2 \theta + 9 \cot^2 \theta} \geq 7$$

Q2 (118)

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Equation of normal of ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ at any point $P (6 \cos \theta, 4 \sin \theta)$ is

$3 \sec \theta x - 2 \operatorname{cosec} \theta y = 10$ this normal is also the normal of the circle passing through the point $(2, 0)$ So,

$6 \sec \theta = 10$ or $\sin \theta = 0$ (Not possible)

$\cos \theta = \frac{3}{5}$ and $\sin \theta = \frac{4}{5}$ so point $P = \left(\frac{18}{5}, \frac{16}{5} \right)$

So the largest radius of circle

$$r = \frac{\sqrt{320}}{5}$$

So the equation of circle $(x-2)^2 + y^2 = \frac{64}{5}$

Passing it through $(1, \alpha)$

Then $\alpha^2 = \frac{59}{5}$

$$10\alpha^2 = 118$$

Q3 (4)

$$y^2 = 3x$$

Tangent P(x_1, y_1) is parallel to $x + 2y = 1$

Then slope at P = $-\frac{1}{2}$

$$2y \frac{dy}{dx} = 3$$

$$\Rightarrow \frac{dy}{dx} = \frac{3}{2y} = -\frac{1}{2}$$

$$\Rightarrow y_1 = -3$$

Coordinates of P(3, -3)

Similarly Q($\frac{4}{\sqrt{3}}, \frac{1}{\sqrt{5}}$), R($-\frac{4}{\sqrt{5}}, \frac{-1}{\sqrt{5}}$)

Area of ΔPQR

$$= \frac{1}{2} \begin{vmatrix} 3 & -3 & 1 \\ \frac{4}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 1 \\ -\frac{4}{\sqrt{5}} & -\frac{1}{\sqrt{5}} & 1 \end{vmatrix}$$

$$= \frac{1}{2} \left[3 \left(\frac{2}{\sqrt{5}} \right) + 3 \left(\frac{8}{\sqrt{5}} \right) + 0 \right] = \frac{30}{2\sqrt{5}} = 3\sqrt{5}$$

Q4 (2)

Equation of normal is

$$2x \sec\theta - by \operatorname{cosec}\theta = 4 - b^2$$

$$\text{Distance from } (0, 0) = \frac{4 - b^2}{\sqrt{4\sec^2\theta + b^2 \operatorname{cosec}^2\theta}}$$

Distance is maximum if

$4\sec^2\theta + b^2 \operatorname{cosec}^2\theta$ is minimum

$$\Rightarrow \tan^2\theta = \frac{b}{2}$$

$$\Rightarrow \frac{4 - b^2}{\sqrt{4 \cdot \frac{b+2}{2} + b^2 \cdot \frac{b+2}{b}}} = 1$$

$$\Rightarrow 4 - b^2 = b + 2 \Rightarrow b = 1 \Rightarrow e = \frac{\sqrt{3}}{2}$$

Q5 (39)

$$\frac{a}{e} = 8 \dots\dots\dots(1) \quad ae = 2 \dots\dots\dots(2)$$

$$8e = \frac{2}{e}$$

$$e^2 = \frac{1}{4} \Rightarrow e = \frac{1}{2}$$

$$a = 4$$

$$b^2 = a^2(1 - e^2)$$

$$= 16 \left(\frac{3}{4} \right) = 12$$

$$\frac{x \cos \theta}{4} + \frac{y \sin \theta}{2\sqrt{3}} = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$P(2\sqrt{3}, \sqrt{3})$$

$$Q\left(\frac{8}{\sqrt{3}}, 0\right)$$

$$(3PQ)^2 = 39$$